

Force

Force (F) is a vector, mass (m) is a scalar.

1 Newton is the force required to give a 1 kg block an acceleration of $1 ms^{-2}$

Types of forces:

Weight - Force exerted on any mass in a gravitational field. Acts through centre of mass.

Tension - Force exerted on an object by strings / chains / pulleys.

Thrust - Force exerted by a spring / rod pushing on a body

Driving force - Force produced by an engine. AKA tractive force.

Resistance to Motion - Acts in opposite dirⁿ to motion. Some types negligible.

Normal Reaction - A force due to 2 bodies being in contact. PERPENDICULAR TO THE LINE OF CONTACT OF THE BODIES.

Resultant of friction + normal force is called THE CONTACT FORCE.

Newton's Laws:

1st Law:

$\sum F = 0 \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} = k$, where k is a constant. EQUILIBRIUM

2nd Law:

$\sum F = m\vec{a}$

$\Rightarrow m = k \Rightarrow \sum F \propto \vec{a}$

$\Rightarrow \sum F = k \Rightarrow \vec{a} \propto \frac{1}{m}$

3rd Law:

$\vec{F}_{A \to B} = -\vec{F}_{B \to A}$

Equal, but opposite, forces exerted on each body

FREEBODY DIAGRAMS

Draw all forces acting on objects

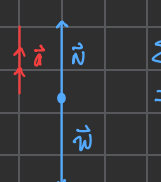
Remember Newton's Third Law

Think Contextually

Which forces are greater

Apparent Weight:

If $\vec{a}$ is in the same dirⁿ as $\vec{N}$, $\Rightarrow \vec{N} > W$. & Vice versa



$\sum F = \vec{N} - \vec{W} = m\vec{a}$

$\Rightarrow \vec{N} = \vec{W} + m\vec{a}$

Where $m\vec{a}$ is an upwards force

$\Rightarrow \vec{N} = \vec{W} + \vec{F}$

Modelling Terms

Inextensible - Constant length

Light - No mass

Negligible - small enough to ignore

Particle - Negligible Dimensions

Smooth - Friction less Surface. Uniform - Constant Throughout.

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$\vec{a}$ is always in the dir^{n} of F_{net}.

Tackling Problems:

Draw clear diagrams

Indicate a positive dir^{n}

Label objects

Show all forces acting on each object.

Make it clear which object you are referring to.

20.6: Connected Objects:

Draw Seperate free diagrams for each object. Sometimes you can also do one for the whole system.

Write equations of motion & quantities for each object